A Minimal Architectural Proposal for Consciousness and Agency on a Shared Neural Backbone

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Abstract

We propose a compact systems-level architecture that separates *shared machinery* from *differential add-ons* to describe the overlap and differences of consciousness and agency. We propose that the shared neural backbone of both comprises (i) a learned generative model, (ii) attention/gain control, and (iii) a short temporal buffer. On top of this, two neural processing stacks branch: a consciousness stack (global broadcast and metacognition) and an agency stack (valuation register and efference cascade).

1 Motivation and scope

Consciousness is used here in a minimalist, operational sense: contents becoming globally available to disparate subsystems and accessible for report [Dehaene, 2014, Dehaene and Naccache, 2001]. **Agency** denotes the capacity to initiate and control actions according to goals, treating systems as if they have beliefs and desires when this makes good predictions for behavior [Dennett, 1987, 1971]. The two frequently co-occur in humans but neither logically entails the other; the architecture below explains the overlap and the dissociations.

The proposal is intentionally model-agnostic. It does not presuppose any specific theory of consciousness or control, and it requires only three building blocks that are broadly accepted in computational neuroscience: predictive modelling [Friston, 2010], attention-like gain control [Sarter et al., 2023], and short-horizon working memory [Miller and Buschman, 2015].

2 Backbone and add-ons: informal specification

Let s_t be latent states, o_t sensations, and a_t actions. The backbone maintains a predictive mapping $p(o_{t+1} \mid s_t)$ and a belief update $q(s_{t+1} \mid s_t, o_{t+1})$ over a *buffer* of length $\tau \in [0,\, 2\,\mathrm{s}]$. A generic attention/gain process α_t modulates precision.

Two add-on stacks branch from this backbone:

- **Consciousness stack**: (C1) *Global broadcast hub*—a routing mechanism that makes selected contents available system-wide; (C2) *Metacognitive monitor*—confidence/error-likelihood tagging of those contents.
- **Agency stack**: (A1) *Valuation register*—a critic-like scalarising function over states and options [Salge et al., 2014]; (A2) *Efference cascade*—an actor-like path from selected policies to motor/effector channels with forward models (efference copy) [Popa and Ebner, 2018].

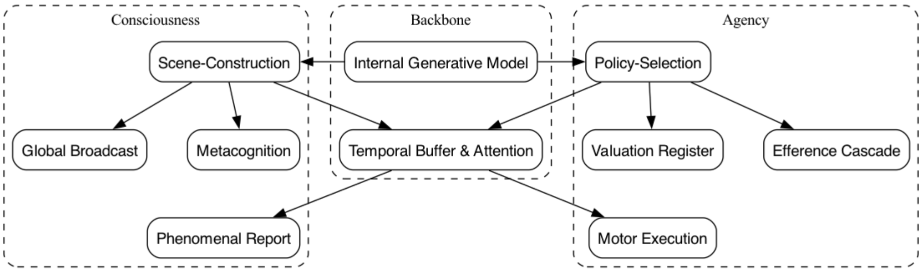

Figure 1: Schematic separation of common machinery (backbone) from differential add-ons for consciousness and agency. The same predictive core serves both stacks; dissociations arise when an add-on fails or is absent.

3 Diagram: shared backbone with differential add-ons

4 Functional roles and minimal predictions

Table 1 states each component's role and gives testable signatures that do not depend on a specific neural implementation.

Component	Functional role	Discriminating operational signatures
Generative model	Maintain a compact, predictive scene state; bind multi-modal inputs over ~ 2 s Burgess et al. [2019], Greff et al. [2019], Locatello et al. [2020]	Violations trigger transient error bursts; imagery substitutes for input with matched dynamics
Attention/gain	Allocate precision to task-relevant channels Locatello et al. [2020]	Manipulating gain selectively boosts reportability or control without changing stimuli
Temporal buffer	Provide a rolling workspace for binding and comparison Kirchhoff et al. [2018a,b]	Lengthening/shortening window shifts integration of rapid sequences (e.g., speech, actions)
Global broadcast (C1)	Make selected contents globally available Kirchhoff et al. [2018a,b]	Loss yields competent behaviour without access/report ("islands"), or vice versa
Metacognition (C2)	Tag contents with confidence/error likelihood Kirchhoff et al. [2018b]	Dissociable from task performance; manipulations shift confidence at fixed accuracy
Valuation register (A1)	Scalarise options via value or cost Kirchhoff et al. [2018b]	Predicts discounting curves and effort allocation; lesions flatten gradients
Efference cascade (A2)	Translate selected policies into actions with forward models Kirchhoff et al. [2018b]	Loss yields intact intent/report but failed execution; efference copy suppresses self-generated reafference
Phenomenal report	Externalisation of selected contents for report or communication Kirchhoff et al. [2018b]	Report channels can be blocked while internal access remains (e.g., output bottlenecks); readouts include verbal report, button press, or eye-tracking
Motor execution	Physical realisation of selected policies via corticobasal and cerebellar loops Kirchhoff et al. [2018b]	TMS/lesion yields intention–execution gaps; efference copy suppresses self-generated reafference, and failures cause sensory instability

Table 1: Roles and generic signatures of backbone and add-ons.

5 Why the architecture explains co-occurrence and dissociation

Because both stacks share the same predictive core, they tend to co-occur in organisms with rich models. Yet each add-on can fail independently:

- (C1) breakdown: intact skilled behaviour with impoverished access/report.

- (A2) breakdown: intact experience of intending with impaired execution.
- (C2) breakdown: normal performance but unreliable confidence control.
- (A1) breakdown: intact perception but apathetic or chaotic policy selection.

These double-dissociations are the architectural expectation, not anomalies.

6 Outputs: Phenomenal report and motor control

Phenomenal report

We use "report" operationally in the sense of Dehaene [Dehaene, 2014]: any externally verifiable channel by which globally broadcast contents are read out (speech, button presses, saccade targets). The report channel is *downstream* of broadcast and metacognition; it can therefore fail independently. Minimal predictions: (i) selective output interference (e.g., speech block) diminishes report without changing internal access; (ii) confidence manipulations shift report thresholds at matched accuracy. This operational approach allows detection of consciousness across species, particularly mammals, through neural patterns comparable to humans during reportable experiences.

Motor control

Motor execution is the physical realisation of selected policies. A forward model (efference copy) predicts reafferent input so that perception remains stable during movement. Minimal predictions: (i) perturbing motor pathways leaves intention judgements intact while degrading execution; (ii) disrupting efference copy induces reafferent confusion (e.g., tickle paradox breakdown).

The agency stack's applicability extends across species where observable intentional behavior can be detected [Dennett, 1987].

Summary. A small set of shared mechanisms plausibly underwrite both consciousness and agency. Two compact add-on stacks account for their frequent coupling and their separable failure modes. This provides a neutral scaffold for cumulative experiments and for comparison with more detailed theories.

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